

CO4-AT2-Q1:

Name: D.Uday Kiran

Reg no: 192425161

Course Code: CSA0652

Course Name: DAA for Complexity Analysis

Autonomous Drone Delivery Routing (Graph + Optimization)

Problem Understanding

Autonomous drone delivery systems require optimized routes to deliver packages quickly while reducing battery consumption. The city environment can be represented as a graph where locations are nodes and paths are edges.

Algorithm Used

Graph-Based Shortest Path Algorithm (Dijkstra Algorithm)

Solution Design

The system represents:

- Delivery locations as vertices
- Roads or flight paths as edges
- Distance or energy cost as edge weights

The algorithm finds the minimum-cost path from the drone source location to all delivery points.

Constraints

- Limited battery capacity
- Weather conditions
- Obstacles and restricted areas
- Multiple delivery locations
- Traffic and route changes

Scalability

For multiple drones:

- Divide delivery areas into regions.
- Use optimized route scheduling.
- Apply parallel processing.

Performance Metrics

- Delivery time
- Battery usage
- Distance travelled
- Number of completed deliveries

Result Analysis

Graph optimization reduces unnecessary travel distance and improves delivery efficiency. The system helps automate logistics operations with faster and safer drone routing.

Conclusion

Graph-based optimization techniques provide efficient route planning for autonomous drones. They improve battery usage, reduce delivery time, and support large-scale logistics automation.